Criminal Legal Supervision and Infant Health

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Abstract: Black-white disparities in infant health remain stark in the United States. Black-white disparities in state-level incarceration rates correlate with the racial inequity in infant health. However, less is known about whether other aspects of the criminal legal system, such as parole and probation, affect racial disparities in these outcomes. Here we examine the association between state-level rates of imprisonment, parole supervision, and probation supervision on Black-white differences in infant mortality. We use two-way fixed effects models with state and year fixed effects to estimate the impact of these state-level characteristics on Black, white, and Black-white differences in infant mortality rates. Preliminary analyses show positive associations between imprisonment rates and Black infant mortality and Black-white inequality, with these associations increasing in both magnitude and significance when including parole and probation rates as predictor variables. The potential role of parole and probation on Black infant health warrants further scrutiny.
(147/150)

INTRODUCTION

Mass incarceration in the United States has numerous well-documented adverse effects on population health, including pronounced impacts on child well-being (Wildeman, Goldman, and Turney 2018; Wildeman and Wang 2017). Focusing on infant health, incarceration is associated with a range of adverse infant health outcomes at the state level, including increased infant mortality rates, lower life expectancy at birth, and low birth weight (Conway 2021; Light and Marshall 2018; Wildeman 2012b, 2012a). At the individual level, parental incarceration during or directly before the prenatal period is associated with increased risk of infant mortality, preterm birth, and low birth weight (Austin, White, and Kim 2022; Lee et al. 2023; Testa et al. 2020; Wildeman 2012a, 2012b).

While there are many plausible mechanisms by which incarceration exposure adversely impacts infant outcomes, few have been empirically examined alongside infant outcomes. Lee and colleagues (2023) found an increased risk of pre-pregnancy hypertension, pre-pregnancy and postpartum depressive symptoms, and smoking during pregnancy among women who were recently exposed to incarceration either directly or through a partner. Additionally, recent work has identified lack of access to prenatal care for incarceration-exposed women as a driver of adverse maternal and infant health outcomes in this population (Bell et al. 2024; Knittel et al. 2022; Steely Smith et al. 2024; Testa et al. 2020). Other hypothesized individual-level pathways include socioeconomic burden, psychosocial stress, and food insecurity (Testa and Jackson 2019; Wildeman 2012b).

Beyond the individual level, mass incarceration can have devastating impacts on community health through increased community levels of distress, economic burden, and policing (Larrabee Sonderlund et al. 2023; Montgomery et al. 2023; Wildeman 2012b). Neighborhood and county level rates of incarceration are associated with increased risk of infant mortality, low birth weight, and preterm birth (Collins et al. 2009; Holaday et al. 2023; Ncube et al. 2016; Sealy-Jefferson et al. 2020). These associations also hold at the state and national levels (Conway 2021; Light and Marshall 2018; Wildeman 2012a, 2012b). Furthermore, resources spent maintaining mass incarceration are withheld from health-promoting welfare programs that could potentially reduce the economic strain associated with incarceration, increase access to healthcare, and mitigate potentially harmful environmental exposures associated with neighborhoods where poverty is concentrated (Wildeman 2012a).

Racial disparities in infant outcomes remain stark in the United States. In 2019, the mortality rate of Black infants was over twice that of white infants (Ely and Driscoll 2021). Individual-level factors do little to explain these Black-white disparities in infant mortality and other infant health outcomes (Matoba and Collins 2017). Instead, scholars have looked to dimensions of structural racism to explain this inequity (Pabayo et al. 2019; Wallace et al. 2017). One such dimension that has been identified as an important driver of Black-white infant health disparities is mass incarceration, which disproportionately impacts Black people in the United States (Boen et al. 2022; Enns et al. 2019; Robey, Massoglia, and Light 2023; Western 2006). Prior literature shows that state-level incarceration rates widen the Black-white gap in life expectancy at birth and that the positive associations between incarceration rate and rate of infant mortality and low-weight births are stronger for Black infants than white infants (Conway 2021; Wildeman 2012a, 2012b; Wildeman and Wang 2017).

While prior literature has focused on the effects of prison and jail incarceration on infant health, less is known about the consequences of other forms of contact with the criminal legal system. Recent work has explored the effects of exposure to police violence and immigration enforcement practices on maternal and infant outcomes, though the literature is relatively sparse (Amuedo-Dorantes, Churchill, and Song 2022; Chegwin et al. 2023; Curtis et al. 2022; Vu 2024). While there is growing interest in examining the impact of these various aspects of the criminal legal system on maternal and infant outcomes, one aspect that, to our knowledge, has never been examined in the context of infant health is community supervision.

The use of both probation and parole is widespread in the criminal legal system with more adults on probation and parole on any given day than adults in prison and jail. For example, there were approximately 3 million adults on probation in 2022 (Kaeble 2024) compared to 1.2 million adults in prison and 660,000 adults in jail (Carson 2023; Zeng 2023). The daily parole population count is smaller in relative size and has seen a significant annual decrease from 2021 to 2022; however, there were still approximately 700,000 adults under parole supervision at the end of 2022 (Kaeble 2024). The extent of both forms of community supervision underscores its potential significance as a health exposure. Research in this area, however, has generally underexplored this pathway, especially when compared to the extensive body of work focused on incarceration (Turney and Wakefield 2019).

Within the limited but growing body of research exploring the connection between community supervision and health outcomes, scholars have found that adults who experienced community supervision had higher risk of chronic conditions, infectious diseases, and substance abuse disorders (Fobian et al. 2018; Hawks et al. 2020; Kilmer and Hughes 2021; Winkelman et al. 2020). With respect to mortality, individuals on probation have an age-adjusted mortality rate that is roughly 2 to 3 times higher than those of individuals incarcerated in jail or prison (Wildeman, Goldman, and Wang 2019). Possible explanations for the observed associations include the inability of the current structure of community supervision to meet the healthcare needs of adults involved in the criminal legal system, the disruptive nature of community supervision, and the constant threat of revocation causing mental strain (Hawks, Horton, and Wang 2022).

What remains unclear in the current literature is the potential impact of community supervision on infant outcomes. It is plausible that, like incarceration, community supervision could act as a stressor for pregnant women by limiting access to prenatal care and imposing additional socioeconomic and psychosocial burdens. Nonetheless, these potential pathways have not been extensively examined in empirical research. Moreover, it is also uncertain whether we can observe any potential association at a broader scale, such as at the state level. This project aims to address these two under-explored questions.

The Present Study

In the present study, we attempt to expand the literature on the role of the criminal legal system in explaining racial infant health disparities by examining the association between yearly, state-level rates of incarceration, probation, and parole and a range of infant health outcomes. Building on Wildeman's (2012b) analyses of the association between yearly imprisonment rates and infant mortality rates at the state level, we extend our analyses to rates of probation and parole from 2000-2018. We also plan to assess additional outcomes aside from infant mortality rate, including fetal mortality rate, race-specific incidence of small for gestational age births, and race-specific incidence of preterm births.

We examine the impact of parole and probation on infant outcomes at the state level to further investigate which aspects of the criminal legal system drive not only adverse infant outcomes but racial disparities in those outcomes. Given the well-documented adverse consequences of incarceration on child well-being and population health at large, we find it important to explore how community supervision may work in tandem with incarceration to shape population health landscapes and reproduce racial health inequities.

ANALYTICAL APPROACH

Data

Data for state-level predictors, outcomes, and covariates were collected from a variety of administrative sources. Variables, descriptive statistics for these variables, specific data sources, and other important information on measures can be found in Table 1. Data were collected for the period 2001-2018.

Variable	Mean	(SD)	Units	Source
Imprisonment Rate	662	(246)	per 100,000 people	U.S. Bureau of Justice Statistics
Parole Rate	306	(260)	per 100,000 people	U.S. Bureau of Justice Statistics
Probation Rate	1,829	(1,071)	per 100,000 people	U.S. Bureau of Justice Statistics
Fetal Death Rate	7.1	(3.2)	per 1,000 live births	National Vital Statistics System
Infant Mortality Rate	7.5	(3.3)	per 1,000 live births	National Vital Statistics System
Preterm Births	13.4	(3.5)	percentage	National Vital Statistics System
Small for Gestational Age	11.4	(3.7)	percentage	National Vital Statistics System
Low Birth Weight	9.68	(3.28)	percentage	National Vital Statistics System
Homicide Rate	4.83	(2.23)	per 100,000 people	U.S. Bureau of Justice Statistics
Violent Crime Rate	397	(149)	per 100,000 people	U.S. Bureau of Justice Statistics
GDP Per Capita	54	(10)	in 2017 dollars	U.S. Bureau of Economic Analysis
Per Capita Health Expenditure	14,199	(3,290)	in current dollars	U.S. Center for Medicare and Medicaid Studies
Average SNAP Benefits Per Household	241	(45)	in current dollars	U.S. Department of Agriculture
Unemployment	5.86	(2.02)	percentage	U.S. Bureau of Labor Statistics

Table 1. State-Level Variables, Descriptive Statistics, and Sources

One important note is that Black infant mortality rates based on a raw count of infant deaths lower than 10 are excluded from analysis. Doing so removes 11 states from the analysis of Black infant mortality rates and Black-white inequality in the infant mortality rate. The remaining states, however, account for the vast majority of Black *and* white births in the given time period.

Methods

We used two-way fixed effects models to estimate the impact of imprisonment, parole, and probation on infant health. State fixed effects control for unobserved differences across states that are constant over time (regional differences, historical differences, etc.); year fixed effects control for unobserved differences over time that are similar across states (national economic trends, national political environment, etc.). We include several contemporaneous controls to adjust for differences that are unique to individual states in individual years. While not an exhaustive list of possible confounders, we believe that our list of covariates balances the need to extract spurious relationships with the drawbacks of over-controlling.

PRELIMINARY RESULTS

For the purposes of analysis, all predictors and covariates were standardized so that coefficients below represent the change in infant mortality rates given a one-standard-deviation increase in the corresponding variable.

Table 2 represents the results from two-way fixed effects models of the association between state-level characteristics and infant mortality rates. The first two models regress *Black* infant mortality rates on state-level characteristics; the third and fourth, *white* infant mortality rates; and the fifth and sixth, *Black-white inequality* in the infant mortality rate. Odd-numbered models are a more straightforward replication Wildeman's (2012b) analyses; even-numbered models add in parole and probation, for the reasons discussed above. Standard errors are clustered at the state level.

These results, although preliminary, allow us to reify and expand Wildeman's (2012b) original analysis. The coefficients in the odd-numbered models tell a similar story today as in 2012: white infant mortality remains relatively unaffected by imprisonment rates, while Black infant mortality and Black-white inequality are both positively associated with imprisonment rates. Further, including parole and probation rates helps to *clarify* the impact of imprisonment rates on infant mortality, increasing both the magnitude and the significance of the coefficient on imprisonment rates. These findings suggest that community supervision may be operating

	Model 1 (B)	Model 2 (B)	Model 3 (W)	Model 4 (W)	Model 5 (B-W)	Model 6 (B-W)
Imprisonment Rate (t-1)	0.58	0.67^{*}	-0.06	-0.06	0.50	0.64^{*}
	(0.31)	(0.33)	(0.09)	(0.08)	(0.29)	(0.30)
Homicide Rate	0.41^{*}	0.41	0.14	0.14	0.26	0.25
	(0.20)	(0.21)	(0.08)	(0.08)	(0.23)	(0.24)
Violent Crime Rate	-0.24	-0.18	0.23^{**}	0.22^{**}	-0.38	-0.29
	(0.26)	(0.27)	(0.07)	(0.07)	(0.27)	(0.28)
State GDP Per Capita	0.95^{**}	0.86^{*}	0.08	0.08	0.70	0.56
	(0.34)	(0.34)	(0.12)	(0.12)	(0.42)	(0.41)
Health Expenditure Per Capita	0.07	0.02	-0.14	-0.13	0.24	0.16
	(0.54)	(0.51)	(0.14)	(0.14)	(0.52)	(0.49)
Average SNAP Benefits Per Household	-1.19^{***}	-1.23^{***}	0.06	0.07	-1.45^{***}	-1.52^{***}
	(0.32)	(0.32)	(0.09)	(0.09)	(0.34)	(0.33)
Unemployment Rate	0.24	0.22	0.06	0.06	0.20	0.17
	(0.18)	(0.18)	(0.05)	(0.05)	(0.21)	(0.21)
Parole Rate		-0.16		0.03		-0.29
		(0.21)		(0.10)		(0.24)
Probation Rate		-0.20		0.01		-0.30
		(0.30)		(0.05)		(0.31)
Num. obs.	599	597	833	831	599	597
Num. groups: state	38	38	49	49	38	38
Num. groups: year	17	17	17	17	17	17
R^2 (full model)	0.70	0.70	0.76	0.76	0.57	0.57
R^2 (proj model)	0.05	0.05	0.03	0.03	0.04	0.05

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 2: Standardized Regression Coe icients from OLS Regression Models with State and Year Fixed Effects Predicting Black and White Infant Mortality Rates and Inequality in the Infant Mortality Rate by State-Level Imprisonment Rates, 2002-2018

differently than imprisonment with respect to infant health. Disentangling this relationship is important to understanding the impact of the criminal legal system.

Planned analyses for this project will expand on these preliminary results in a few key ways. First, they will include indices of poverty, high-risk birth, and population heterogeneity, to better account for possible spuriousness. Second, they will test if autoregressive standard errors, which may better represent the serially correlated structure of the data, change the estimates substantially. Finally, they will focus on other key indicators of fetal and infant health, such as fetal death rates and percent preterm births. Taken together, we think that these analyses will achieve two important goals: first, bringing Wildeman's (2012b) analysis into the present, and second, providing a more complete understanding of the impact of the criminal legal system on infant health.

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